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Task Description:	Using very accurate optical imaging techniques and particle image velocimetry, we characterize the shape of the fluid/fluid interface and the flow fields in a microscopic region near moving contact lines. From these images, we test models of the hydrodyna mics in regions near the contact line where viscous forces dominate the fluid flow and in regions even closer to the contact line where a unique hydrodynamic regime must govern the fluid motion. By making systematic changes in the materials system under examination, we seek the physical origins of these unique hydrodynamics. Low gravity studies are essential to this program; bouyancy flows that would otherwise mask the effects of capillary and wetting forces are eliminated. The study will characterize the unique hydrodynamics occurring in a microscopic region near moving contact lines and that controls macroscopic spreading of fluids on solid surfaces. The task significance is to (1) establish the correct method for measuring dynamic contact angle boundary condition, (2) provide a proper basis for predictive models of spreading, (3) test models of the hydrodynamics near, but not at, moving		

	contact lines where viscous forces are dominant, and (4) identify the molecular-scale mechanisms governing the unique hydrodynamic regimes at moving contact lines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions adn elastic fluids; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating or printing, which employ dynamic wetting by surfactant solutions and non-Newtonian fluds.
	During the past year, we have investigated dynamic wetting by two classes of complex fluids: surfactant solutions and polymer melts. We have found complex behavior when a solid surface is pulled from a bath of surfactant solution. At some critical velocity, a film of fluid is entrained onto the solid surface as if emerges from the bath. We have found an abrupt transition in the critical velocity at a characteristic surfactant concentration. This transition appears in a wide range of systems. Below that film pulling velocity, we find the contact lines of surfactant solutions do not remain smooth nor retreat by steady motion across the surface. Rather, the contact line is mostly flat and moving steadily but micron scale fluctuations occur. These fluctuations suggest that, sporadically and locally, the surfactant molecules cannot restructure fast enough as they pass through the contact line (the edge of the fluid on the surface) to form an equilibrium structure. Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating, which employ dynamic wetting by surfactant solutions. Non-Newtonian fluids are employed in many technologies, yet their wetting behavior has received far less attention than Newtonian fluids. We have found that two room temperature polymer melts, polyisobutylene and polystyrene, do not exhibit the same dynamic wetting behavior as another polymer melt, polydimethylsiloxane. All three fluids appear Newtonian in standard rotational rheological measurements, but only the latter shows wetting behavior explained by a model which assumes Newtonian behavior of the fluid. The polyisobutylene and polystyrene show behavior that is indicative of weak elasticity in the fluid. These experiments show that dynamic wetting is more sensitive to this elasticity than standard rheometric techniques. Probing fluids with strong elasticity, we have found that a series of polymer mixtures, Boger fluids, deviate strongly from Newtonian wetting behavior and may exhibit a fluid instability near the moving contact line. This research has direct application in developing well-founded avenues to design of the many technologies (e.g. painting, ink jet printing) that employ non-Newtonian fluids. Dynamic Wetting of Surfactant Solutions Universal Transition in the Critical Velocity for Film Pulling A film is entrained onto a solid substrate when it is withdrawn from a bulk fluid above a critical velocity, Uc. For surfactant solutions, this critical velocity exhibits a large and abrupt change between 0.3 x critical micelle concentration (CMC) and 0.6 CMC. Below 0.3CMC the critical velocity for film entrainment is greater than 2 cm/s. Above 0.6CMC the critical velocity of film entrainment abruptly jumps to about 100 microm/s. If we convert the critical velocity to a critical capillary number, (Cac=muUc/gamma where mu is viscosity of the fluid, Uc is the speed at which the substrate is withdrawn from the fluid bath and gamma is the liquid/vapor surface tension), this transition is independent of surfactant charge, surfactant size and the substrate surface energy. Other wetting properties such as autophobing and dendritic spreading also show a transition at ~0.5CMC. However, none of the interfacial properties such as liquid-vapor surface tension or the surface excess concentration at the solid/liquid or liquid/vapor interfaces show such a jump between 0.3 CMC and 0.6 CMC. We have found this transition in critical velocity for a wide range of systems including: a cationic surfactant (cetyltrimethylammonium bromide, CTAB), CTAB with 10mM KBr, and a nonionic surfactant (octaethylene glycol monododecyl ether, C12E8) on both charged, hydrophilic (bare SiO2) and methylated, hydrophobic (octadecyltrichlorosilane, OTS, coated SiO2) surfaces. For pure viscous fluids, models suggest that the critical capillary number should be roughly proportional to the third power of the static contact angle. However in our systems, not only does the static contact angle not make an abrupt jump between 0.3 CMC and 0.6 CMC; but no correlation has been identified between the static contact angle and the critical capillary number. Fluctuations in Receding Contact Lines of Surfactant Solutions For several years, our group has probed the unsteady motion of advancing contact line of surfactant solutions that arises from dynamic surfactant self-assembly ahead of the contact line. Our recent efforts have shown that receding contact lines of surfactant solutions are also inherently unsteady. Contact line fluctuations have been observed when a substrate is withdrawn from a surfactant solution at velocities as slow as two microns/sec and up to the velocity of film entrainment. As the surface is withdrawn, while much of the contact line remains flat and steady, small (laterally ~10micron) distortions of the contact line are seen. These distortions appear to be pinned to the surface as it emerges from the fluid bath; and at some height (again ~10microns), they break loose from the surface and rejoin the flat contact line. We have observed these fluctuations for a wide variety of systems including: ionic surfactants, ionic surfactant solutions with salt added to screen the head group/surface interaction, and nonionic surfactants on both ionic oxide and nonpolar surfaces. The frequency of these fluctuations increases as the interaction between the head group and the surface weakens and as the velocity of the surface increases. For strong interactions (e.g. CTAB on silica) the contact line is virtually flat while the same surfactant on a surface coated with octadecyltrichlorosilane shows frequent fluctuations at the same withdrawal speed. These distortions show all the characteristics of quasi-static distortions of a contact line that are induced by chemical heterogeneity on a surface. However, analysis of the frequency of the fluctuations shows that they do not arise from nonuniformities in the underling surface. Rather, the fluctuations we observe likely originate in localized failure of the surfactant to uniformly self-assemble as it passes through the contact line, thus forming a local region ahead of the contact line that is more hydrophilic than the majority of the contact line, where a more hydrophobic self-assembly has formed. AFM studies confirm this picture, showing regions of excess surfactant deposition on the surface of the same size and occurring with he same frequency as the fluctuations seen in the contact line. Thus, these fluctuations probe a fundamental relaxation time of the surfactant-assembly as it passes through a contact line. Our studies have shown that these fluctuations do not lead to film pulling. First. while film pulling requires the

Task Progress:

macroscopic contact angle to become 0°, the fluctuations do not drive the surface to local contact angle of 0°. Second, as the critical velocity for film pulling is approached, the localized fluctuations do not increase frequency so much that they fill the contact line. Thus, we see the fluctuations are not the direct cause of the film pulling which is a hydrodynamic phenomenon.

Future Work

1. We will probe details of the liquid/vapor interface shape near the critical velocity for non-volatile, pure fluids (polydimethylsiloxanes) and surfactant solutions to study the underlying hydrodynamics of the onset of film pulling. The first system will provide a simpler case; while in the second case, we must consider possible effects due to Marangoni flows.

2. When surfactants are present, a dividing streamline may appear in the fluid near the contact line. We have evidence that this is indeed occurring. Using surface particle image velocimetry, we will now measure the dependence of the location of the dividing streamline on contact line speed, surfactant concentration, and surfactant/surface interaction.

3. Using optical ellipsometry, optical reflectometry, and x-ray reflectometry, we will quantitatively measure the mass transfer of surfactant among all three interfaces and the bulk solution for a receding contact line.

4. We will probe the influence of Marangoni flows on the shape of films of surfactant solutions entrained on a surface as it is pulled from the fluid bath above the critical velocity.

Technological Impact

Dynamic wetting by surfactant solutions occurs in many technologies (e.g., cleaning and coating processes, oil recovery from reservoirs). Control of contact line hydrodynamics plays a critical role in optimizing such processes. Our research will elucidate these fundamental hydrodynamics, especially the origins of film pulling and deposition of material out of the moving contact line.

Dynamic Wetting of Polymer Melts

Non-Newtonian Behavior in Simple Polymer Melts

The hydrodynamics near moving contact lines of two room temperature polymer melts, polyisobutylene and polystyrene, are different from those of a third polymer melt, polydimethylsiloxane. While all three fluids exhibit Newtonian behavior in rotational rheological measurements, a model of the hydrodynamics near moving contact lines which assumes Newtonian behavior of the fluid accurately describes the interface shape of a variety of polydimethylsiloxane fluids but fails to describe the interface deformation by viscous forces in polyisobutylene and polystyrene. Polyisobutylene and polystyrene showed more viscous bending near a moving contact line than predicted by the Newtonian model. The magnitude of the deviations from the model and the distance along the liquid-vapor interface over which they are seen increase with increasing capillary number. A fluid whose non-Newonian behavior is dominated by shear–thinning, an aqueous solution of xanthan gum, shows less curvature than predicted by the model which describes polydimethylsiloxane, while a fluid whose non-Newtonian behavior are dominated by elasticity, SMX (a Boger fluid composed of a small amount of high molecular weight polystyrene in ologomeric polystyrene) shows more curvature than the model. Thus, we concluded that the wetting behaviors of polyisobutylene and polystyrene are influenced by weak elasticity in these low molecular weight melts and that dynamic wetting is more sensitive to this elasticity than standard rheometric techniques.

Dynamic Wetting in Boger Fluids

To probe dynamic wetting by fluids whose rheology is dominated by elastic behavior, we are measuring the dynamic wetting of a set of model Boger fluids, the SMX fluids mentioned above. These fluids show strong elasticity with a shear viscosity that does not vary with shear rate. The series consists of three fluids with increasing elasticity. As expected, none of these fluids show dynamic wetting described by a model which as successfully described the wetting of Newtonian fluids. Each shows increased viscous bending near the contact line. In one of the fluids, the interface shape near the contact line is anomalous, exhibiting a strong break in the interface curvature about ~100 microm from the contact line. This break may suggest a fluid instability arising in the nonviscometric flows near the contact line. Characterization of this break is critical to controlling the wetting of elastic fluids in applications, since it will strongly impact the macroscopic spreading of elastic fluids,

Unusual Static Wetting Behavior of Polymer Melts

Polyisobutylene and polystyrene show non-zero static contact angles on a bare silica surface. Energy considerations (the critical surface energy of silica is far greater then the interfacial tension of these polymers) suggest that the fluids should wet the surface. Polydimthylsiloxane does have a 0° contact angle on silica showing that the effect seen is not due to the polymeric nature of these fluids. We hypothesize that a monolayer of the polymer moves across the surface ahead of the contact line; and in that layer, the forces between the silica and the molecules orient the molecules, with less polar groups pointed away form the surface. This then may produce a low energy surface that is not wet by the polymeric fluid itself. Such “autophobing” as been frequently often seen in surfactant solutions but has not been identified in a polymeric system. Our efforts to detect the monolayer ahead of the contact line suggest it is less than 20?m in extent, in contrast to surfactants where this monolayer can be millimeters long. This may arise from the slow diffusive growth of the monolayer of polymeric molecules. The small extent of the film in the polymer case emphasizes how small a structure ahead of the contact line is needed to alter wetting behavior.

Future Work

1. Using both our telemicroscopy techniques to measure liquid/vapor interface shapes and particle image velocimetry, we will determine both the origin of the apparent interface instability observed in one of our Boger fluids and the parameter range over which it occurs. 2. We will make careful measurements of the curvature of the interface shape of the SMX near a moving contact line. We also will make measurements of the relaxation of the interfaces of Boger fluids (e.g. measuring drop spreading). Guided by these results and using an appropriate constitutive equation, we will attempt to model the hydrodynamics near contact lines of elastic fluids.

3.We will prepare and examine polyisobutylene based Boger fluids to determine whether the wetting behavior exhibited by the polystyrene based Boger fluids is unique.

	4. By observing the contact angle and film formation of receding menisci of polyisobutylene and polystyrene, we will determine if precursing monolayers are responsible for the non-zero advancing static contact angles we have observed. Technological Impact These studies will develop a complete picture of the hydrodynamics that control wetting of non-Newtonian fluids. They will establish the principles for design of the many technological processes involving wetting by non-Newtonian fluids.
Bibliography Type:	Description: (Last Updated: 07/02/2007)
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Articles in Peer-reviewed Journals	G.K. Seevaratnam, L. Walker, E. Rame, S. Garoff "Wetting by Simple Room Temperature Polymer Melts: Deviations from Newtonian Behavior" J. Coll. Interf. Sci. 284 (1): 265 , Jan-2005
Presentation	E. Rame "The Influence of Dynamic Contact Angle on wetting Dynamic" ALA Lab Automation Jun-2004
Presentation	G. Seevaratnam, L.M. Walker, S. Garoff and E. Rame "Wetting by Simple Room Temperature Polymer Melts: Deviations from Newtonian Behavior" 78th ACS Colloids and Surface Science Symposium Jun-2004
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Presentation	Gita Seevaratnam, Stephen Garoff , Lynn M. Walker, Enrique Rame "Influence of Non-Newtonian Behavior on the Dynamic Interface Shapes of Polymer Melts, Solutions and Boger Fluids" 79th ACS Colloid and Surface Science Symposium Jun-2005